

Supplementary information for

Accelerating physics-informed neural networks for full waveform inversion using a hybrid quantum-classical finite-basis architecture

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Supplementary Note: Quantum background

For completeness, we summarize the basic quantum-computing concepts used by the hybrid architecture. A single qubit is a two-level quantum system with state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (\text{S1})$$

where $\alpha, \beta \in \mathbb{C}$ are probability amplitudes. An n -qubit register is represented by a normalized statevector in $\mathbb{C}^{2^n}$,

$$|\psi\rangle = \sum_{k=0}^{2^n-1} \alpha_k |k\rangle, \quad \sum_{k=0}^{2^n-1} |\alpha_k|^2 = 1. \quad (\text{S2})$$

The parameterized single-qubit rotations used in this work are

$$R_k(\theta) = e^{-i\theta \sigma_k/2}, \quad k \in \{x, y, z\}, \quad (\text{S3})$$

with explicit matrices

$$R_x(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}, \quad R_y(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}, \quad R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}. \quad (\text{S4})$$

The composition $R_z(\theta_3)R_y(\theta_2)R_x(\theta_1)$ spans $\text{SU}(2)$ and can prepare any single-qubit pure state up to a physically irrelevant global phase.

Entanglement between neighbouring qubits is introduced by the controlled-NOT (CNOT) gate,

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (\text{S5})$$

which flips the target qubit if and only if the control qubit is in state $|1\rangle$.

Supplementary Note: Fourier structure of the angle-encoded PQC

This note derives the finite-frequency structure of the PQC output used in the Discussion, specializing the general result of Schuld, Sweke, and Meyer (2021) to the angle encoding adopted in this work.

Consider first a single encoded scalar s with Hermitian generator G and encoding unitary $U_{\text{enc}}(s) = e^{-i\beta s G}$, acting on the reference state $|\psi_0\rangle = |0\rangle^{\otimes n}$. Because the variational block $V(\boldsymbol{\theta}_q)$ is applied after the encoding, the averaged measurement observable $O = \frac{1}{n} \sum_{i=0}^{n-1} Z_i$ can be folded into the Heisenberg-picture operator $M(\boldsymbol{\theta}_q) = V(\boldsymbol{\theta}_q)^\dagger O V(\boldsymbol{\theta}_q)$, so the measured output is

$$f(s) = \langle \psi_0 | U_{\text{enc}}(s)^\dagger M(\boldsymbol{\theta}_q) U_{\text{enc}}(s) | \psi_0 \rangle. \quad (\text{S6})$$

Writing the spectral decomposition $G = \sum_a \lambda_a \Pi_a$ gives $U_{\text{enc}}(s) = \sum_a e^{-i\beta s \lambda_a} \Pi_a$, and therefore

$$f(s) = \sum_{a,b} e^{i\beta s (\lambda_a - \lambda_b)} \langle \psi_0 | \Pi_a M(\boldsymbol{\theta}_q) \Pi_b | \psi_0 \rangle, \quad (\text{S7})$$

so the accessible frequency indices are the pairwise eigenvalue differences, $\Omega \subseteq \{\lambda_a - \lambda_b\}$. For the $R_y(\beta s)$ encoding used here, $G = Y/2$ has eigenvalues $\pm \frac{1}{2}$, so the differences are $0, \pm 1$ and

$$f(s) = A + B \cos(\beta s) + C \sin(\beta s). \quad (\text{S8})$$

The trainable angles change A , B , and C but cannot create frequencies outside this set.

For the full register, each coordinate z_i is encoded on its own qubit through $R_y(\beta_i z_i)$, so the encoding generators $\{Y_i/2\}$ act on disjoint tensor factors and mutually commute. The encoding unitary then factorizes as $\prod_i e^{-i\beta_i z_i Y_i/2}$, the per-coordinate spectra combine additively, and the accessible index set becomes $\Omega \subseteq \{-1, 0, 1\}^n$, with physical frequencies $\boldsymbol{\beta} \odot \boldsymbol{\omega}$. This is the multivariate Fourier form quoted in the Discussion. The entangling CNOT layers and variational angles determine which coefficients $c_{\boldsymbol{\omega}}$ are nonzero, in particular populating cross-coordinate indices, but they do not enlarge Ω .

Supplementary Figures

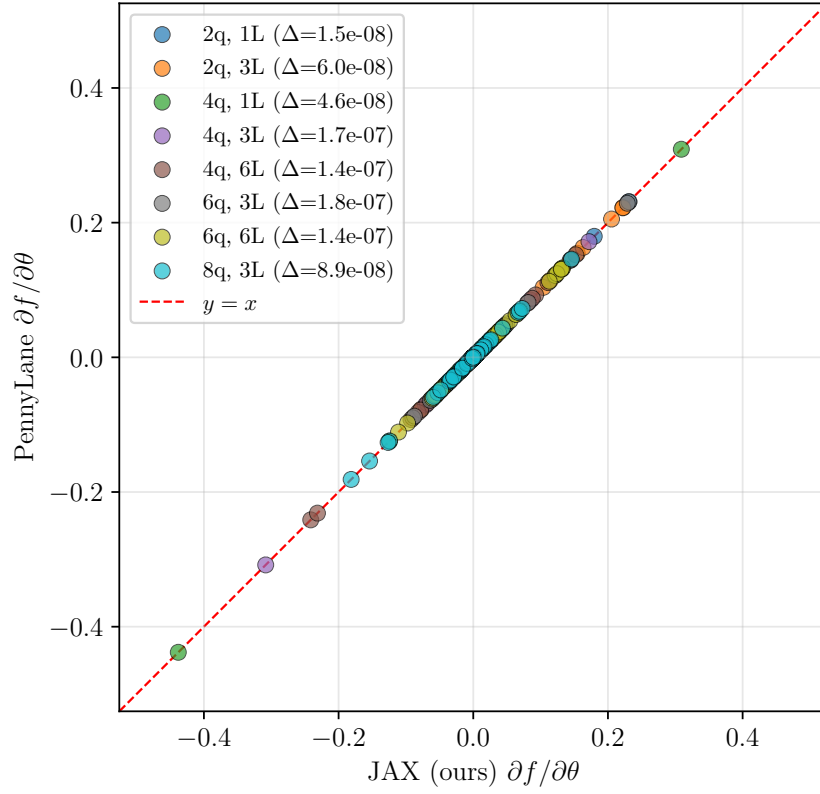


Figure S1: Comparison of parameter gradients $\partial f / \partial \theta_q$ computed by our JAX statevector implementation and PennyLane’s `default.qubit` backend across eight circuit configurations (2–8 qubits, 1–6 layers). All values fall on the identity line $y = x$ with maximum absolute deviation $\Delta < 2 \times 10^{-7}$, confirming numerical equivalence of the two implementations.

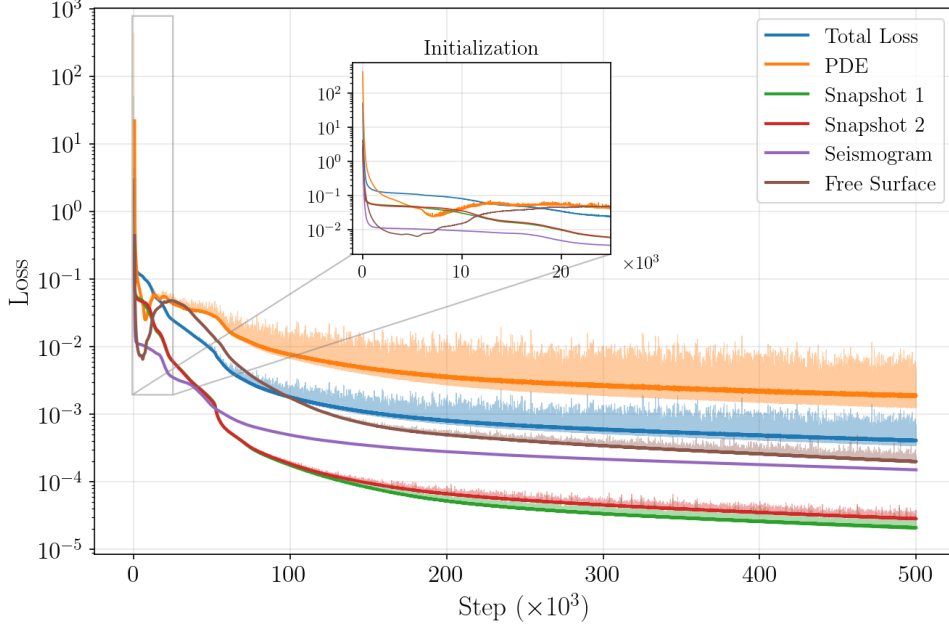


Figure S2: Training loss decomposition for the classical FBPINN baseline on the anomaly benchmark: total loss, PDE residual, displacement-snapshot initial conditions (two components), seismogram misfit, and free-surface boundary condition, plotted versus training iteration. Inset: early-iteration behaviour showing rapid initial loss reduction.

Supplementary Tables

Table S1: Full hyperparameter variant sweep for the anomaly benchmark. $\mathcal{Q}(n_q^{n_L})$ denotes a parameterized quantum circuit with n_q qubits and n_L layers. Values marked “—” are identical to the `classical` baseline. Subdomains column shows $\phi(\cdot)$ for wavefield decomposition.

Model	ϕ network $(x, z, t) \rightarrow \phi$	α network $(x, z) \rightarrow \alpha$	LR	Subs	Overlap	λ_{PDE}	λ_{IC}	λ_{data}	λ_{BC}	# Pars
<code>classical</code>	$\mathcal{N}: 32 \times 2 \rightarrow 16 \times 2$	$\mathcal{N}: 20 \times 5$	10^{-4}	$\phi(5 \times 3 \times 5)$	0.35	0.1	1.0	1.0	0.1	151,836
<i>ϕ-network size variants</i>										
<code>phi16x2</code>	$\mathcal{N}: 16 \times 2$	—	—	—	—	—	—	—	—	25,836
<code>phi64x3</code>	$\mathcal{N}: 64 \times 3$	—	—	—	—	—	—	—	—	649,836
<i>ϕ subdomain decomposition variants</i>										
<code>sub3x2x3</code>	—	—	—	$\phi(3 \times 2 \times 3)$	—	—	—	—	—	37,779
<code>sub8x4x8</code>	—	—	—	$\phi(8 \times 4 \times 8)$	—	—	—	—	—	514,017
<i>α-network size variants</i>										
<code>alpha10x2</code>	—	$\mathcal{N}: 10 \times 2$	—	—	—	—	—	—	—	150,226
<code>alpha10x3</code>	—	$\mathcal{N}: 10 \times 3$	—	—	—	—	—	—	—	150,336
<code>alpha20x3</code>	—	$\mathcal{N}: 20 \times 3$	—	—	—	—	—	—	—	150,996
<code>alpha40x3</code>	—	$\mathcal{N}: 40 \times 3$	—	—	—	—	—	—	—	153,516
<code>alpha40x5</code>	—	$\mathcal{N}: 40 \times 5$	—	—	—	—	—	—	—	156,796
<code>alpha60x3</code>	—	$\mathcal{N}: 60 \times 3$	—	—	—	—	—	—	—	157,636
<i>Training hyperparameter variants</i>										
<code>lr5e4</code>	—	—	5×10^{-4}	—	—	—	—	—	—	—
<code>lr5e5</code>	—	—	5×10^{-5}	—	—	—	—	—	—	—
<code>overlap50</code>	—	—	—	—	0.50	—	—	—	—	—
<code>wpde01_wbc01</code>	—	—	—	—	—	0.01	—	—	0.01	—
<code>wpde1_wbc1</code>	—	—	—	—	—	1.0	—	—	1.0	—
<i>Quantum hybrid</i>										
<code>quantum</code>	$\mathcal{N}: 32 \times 2$ $\rightarrow \mathcal{Q}: 4 \times 2$	$\mathcal{N}: 20 \times 3$ $\rightarrow \mathcal{Q}: 4 \times 3$	—	—	—	—	—	—	—	101,812